

The Anomaly Detection Algorithm 🔍

We can now build the full algorithm using the concept of the Gaussian distribution.

The Model: Multiplying Probabilities

We have a training set where each example x has n features $(x_1, \dots, x_n)$.

- *Example:* Aircraft engine with 2 features (Heat, Vibration). $n = 2$.

To estimate the total probability $p(x)$ of an entire vector of features, we assume the features are statistically independent (though the algorithm works well even if they aren't) and **multiply the probabilities of each individual feature**.

$$p(x) = p(x_1; \mu_1, \sigma_1^2) \times p(x_2; \mu_2, \sigma_2^2) \times \dots \times p(x_n; \mu_n, \sigma_n^2)$$

This can be written compactly as:

$$p(x) = \prod_{j=1}^n p(x_j; \mu_j, \sigma_j^2)$$

- $\prod$ is the product symbol (like $\sum$ for summation), meaning "multiply all these terms together."
 - Each feature j has its own mean μ_j and variance σ_j^2 .
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The Algorithm Steps

1. Choose Features

Select features x_i that you believe are indicative of anomalous behavior.

2. Fit Parameters (μ and σ^2)

Calculate the mean and variance for **each feature j** using the training set.

- $\mu_j = \frac{1}{m} \sum_{i=1}^m x_j^{(i)}$ (Average of feature j)
- $\sigma_j^2 = \frac{1}{m} \sum_{i=1}^m (x_j^{(i)} - \mu_j)^2$ (Variance of feature j)

3. Compute $p(x)$ for New Examples

Given a new example x (e.g., a new engine), calculate its total probability:

$$p(x) = \prod_{j=1}^n \frac{1}{\sqrt{2\pi\sigma_j^2}} \exp\left(-\frac{(x_j - \mu_j)^2}{2\sigma_j^2}\right)$$

4. Evaluation (Flag Anomaly)

Compare the calculated $p(x)$ to a threshold ϵ .

- If $p(x) < \epsilon$: Flag as **ANOMALY**.
 - If $p(x) \geq \epsilon$: Consider it **NORMAL**.
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Intuition

- The algorithm fits a bell curve to *each* feature independently.
 - If a new example has even **one** feature that is very far from the mean (e.g., excessively high heat), the probability $p(x_j)$ for that specific feature will be tiny.
 - Because we are multiplying, **multiplying by a tiny number makes the total product tiny**.
 - Therefore, if any single feature is anomalous, the total $p(x)$ will drop below ϵ , and the system will flag it.
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Example Visualization

- Imagine data with 2 features.
 - x_1 : Heat (Mean=5, Sigma=2)
 - x_2 : Vibration (Mean=3, Sigma=1)
- Multiplying these two distributions creates a 3D "hill" of probability.
 - The peak (highest probability) is in the center, where the means are.
 - The edges (low probability) are where anomalies live.
- **Test 1 (Center):** $p(x) = 0.4$ (Normal)
- **Test 2 (Far Edge):** $p(x) = 0.0021$ (Anomaly)